Position: Rethinking Post-Hoc Search-Based Neural Approaches for Solving Large-Scale Traveling Salesman Problems

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Combinatorial optimization already impacts real systems at scale

- Routing, logistics, and network design drive global efficiency and sustainability.
- The Traveling Salesman Problem (TSP) is a cornerstone with real-world impact.
- ML is increasingly combined with operations research to tackle scale.

Large-scale TSP demands reliable solutions under tight compute budgets

- Goal: high-quality tours for 500–10,000 nodes under realistic resources.
- Trend: heatmap-guided MCTS uses ML-predicted edge priors to refine tours.
- Question: does this guidance add value over strong heuristics?

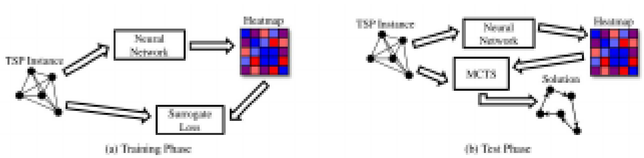


Figure: Heatmap-guided MCTS phases.

Surrogate objectives and unfair budgets undermine heatmap-guided MCTS

- Training losses often misalign with the search used at inference.
- Resource fairness is unclear versus strong OR baselines like LKH-3.
- Uniform or near-zero heatmaps can perform similarly in practice.

A simple, aligned baseline and fair scoring overturn prior claims

- Provide a systematic theoretical and empirical critique of heatmap-guided MCTS.
- Introduce SoftDist: a supervision-free, objective-aligned heatmap with negligible latency.
- Define a resource-fair comparison protocol and gap-ratio score to benchmark against LKH-3.
- Show that under matched budgets, LKH-3 remains strongest; SoftDist matches prior ML quality at far lower cost.

MCTS uses heatmaps only as weak priors for local k-opt search

- An ML heatmap provides edge-inclusion priors; MCTS biases local improvements accordingly.
- Default MCTS parameters are shared across heatmap sources for fairness and a balanced search.
- Search gains are backpropagated in the tree; the heatmap influences but does not dictate moves.

A one-parameter distance-based heatmap rivals complex ML models

- Compute $\Phi_{ij} = \mathrm{softmax}_j(-d_{ij}/\tau)$ so shorter edges get higher probability.
- A single temperature controls exploration versus greediness; no labels and negligible cost.
- In practice, this minimal prior competes with learned heatmaps at a fraction of the latency.

Optimizing τ on the actual search loop closes the train–test gap

- Select the temperature per scale by grid-searching directly on the same MCTS used at test time.
- This aligns the objective with the deployed search, avoiding training–inference mismatch.
- A short, coarse grid search finds near-optimal settings per scale; sensitivity is mild except at very small temperatures on very large instances.

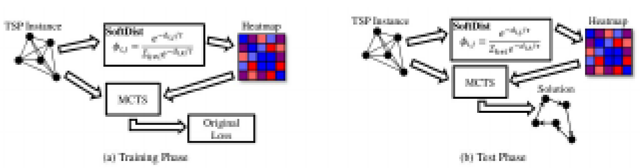


Figure: SoftDist-guided MCTS phases with temperature tuning in the loop.

Matched hardware and a gap-ratio metric enable fair comparisons

- Gap versus a strong baseline is the relative excess tour length.
- Score compares MCTS and LKH-3 by the ratio of their gaps; higher means MCTS is closer to the baseline.
- Setup: synthetic Euclidean TSP-500/1000/10000; shared MCTS configs and CPU; SoftDist generation on GPU adds negligible latency.
- Interpretation: 100% means parity; below 100% favors LKH-3; above 100% favors MCTS.

SoftDist matches quality at negligible latency

- X-axis sweeps heatmap generation time; SoftDist sits near zero while maintaining strong tour quality.
- Headline: comparable or better quality than prior ML heatmaps with orders-of-magnitude lower latency.

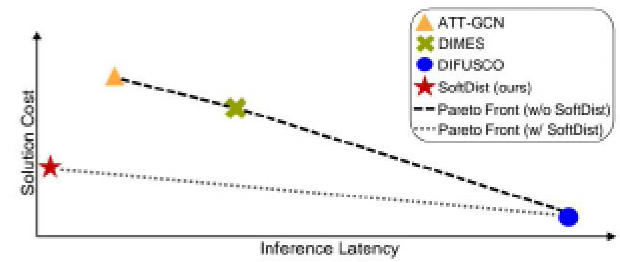


Figure: Lower inference latency (x) and lower solution cost (y) are better; SoftDist is near-zero latency with competitive quality.

Across scales, SoftDist is competitive; LKH-3 remains strongest

- SoftDist+MCTS achieves small gaps across 500/1000/10000 nodes with negligible heatmap time.
- DIFUSCO attains similar quality but requires heavy supervision and long inference time.
- Strong OR (LKH-3) is still the most competitive in both quality and efficiency.

Table: Table 1. Results on large-scale TSP problems.

Method	Type	TSP-500			TSP-1000			TSP-10000		
		Length ↓	Gap ↓	Time ↓	Length ↓	Gap ↓	Time ↓	Length ↓	Gap ↓	Time ↓
CONCORDE	OR(EXACT)	16.55*	—	37.66M	23.12*	—	6.65H	N/A	N/A	N/A
GUROBI	OR(EXACT)	16.55	0.00%	45.63H	N/A	N/A	N/A	N/A	N/A	N/A
LKH-3 (DEFAULT)	OR	16.55	0.00%	46.28M	23.12	0.00%	2.57H	71.78*	—	8.8H
LKH-3 (LESS TRAILS)	OR	16.55	0.00%	3.03M	23.12	0.00%	7.73M	71.79	—	51.27M
NEAREST INSERTION	OR	20.62	24.59%	0S	28.96	25.26%	0S	90.51	26.11%	6S
RANDOM INSERTION	OR	18.57	12.21%	0S	26.12	12.98%	0S	81.85	14.04%	4S
FARTHEST INSERTION	OR	18.30	10.57%	0S	25.72	11.25%	0S	80.59	12.29%	6S
EAN	RL+S	28.63	73.03%	20.18M	50.30	117.59%	37.07M	N/A	N/A	N/A
EAN	RL+S+2-OPT	23.75	43.57%	57.76M	47.73	106.46%	5.39H	N/A	N/A	N/A
AM	RL+S	22.64	36.84%	15.64M	42.80	85.15%	63.97M	431.58	501.27%	12.63M
AM	RL+G	20.02	20.99%	1.51M	31.15	34.75%	3.18M	141.68	97.39%	5.99M
AM	RL+BS	19.53	18.03%	21.99M	29.90	29.23%	1.64H	129.40	80.28%	1.81H
GCN	SL+G	29.72	79.61%	6.67M	48.62	110.29%	28.52M	N/A	N/A	N/A
GCN	SL+BS	30.37	83.55%	38.02M	51.26	121.73%	51.67M	N/A	N/A	N/A
POMO+EAS-EMB	RL+AS	19.24	16.25%	12.80H	N/A	N/A	N/A	N/A	N/A	N/A
POMO+EAS-LAY	RL+AS	19.35	16.92%	16.19H	N/A	N/A	N/A	N/A	N/A	N/A
POMO+EAS-TAB	RL+AS	24.54	48.22%	11.61H	49.56	114.36%	63.45H	N/A	N/A	N/A
DIFUSCO‡	SL+MCTS	16.63	0.51%	3.61M+1.67M	23.39	1.18%	11.86M+3.34M	73.76	2.77%	28.51M+16.87M
ATT-GCN†	SL+MCTS	16.82	1.64%	0.52M+1.67M	23.67	2.37%	0.73M+3.34M	74.50	3.80%	4.16M+16.77M
DIMES†	RL+MCTS	16.84	1.77%	0.97M+1.67M	23.68	2.44%	2.08M+3.34M	74.10	3.21%	4.65M+16.77M

Even with longer budgets, MCTS trails LKH-3 across scales

- X-axis sweeps wall-clock time; LKH-3 improves faster and sustains a lead.
- Headline: longer budgets do not close the gap; SoftDist is best among MCTS but still behind LKH-3.

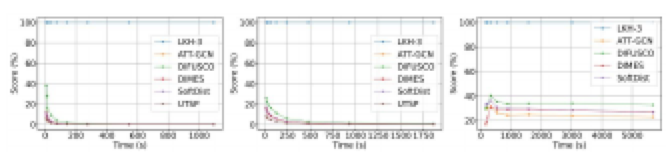
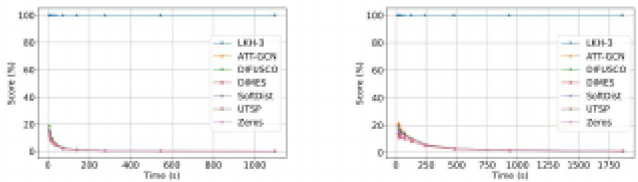


Figure: Performance of MCTS under default settings for TSP-500, TSP-1000, and TSP-10000.

Tuning MCTS barely changes the story across heatmaps

- X-axis sweeps time under alternative parameters; relative ordering among heatmaps changes little.
- Headline: SoftDist remains competitive without latency; DIFUSCO is strong but slow; a zero heatmap is close to learned heatmaps.



(a) TSP-500 with different MCTS settings.

(b) TSP-1000 with different MCTS settings.

Figure: Under alternative MCTS parameters (Min et al., 2023), relative ordering across heatmaps barely changes.

Under matched resources, MCTS efficiency lags far behind LKH-3

- Gap-ratio Scores are far below LKH-3 on 500/1000 nodes, and remain low at 10k nodes.
- More supervision and heavier GPUs do not guarantee practical wins.

Table: Table 2. Resource consumption and Score comparison of various methods on TSPs.

Method	Supervision	Hardware	TSP-500	TSP-1000	TSP-10000
ATT-GCN†	✓	GTX 1080 TI GPU	0.74%	3.87%	24.66%
DIMES†	×	NVIDIA P100 GPU	0.68%	3.75%	28.99%
UTSP†	×	NVIDIA V100 GPU	0.35%	2.08%	—
DIFUSCO#	✓	8×NVIDIA V100 GPUS	2.39%	7.78%	33.82%
SOFTDIST	×	NVIDIA A100 GPU	0.84%	4.17%	29.88%

Near-zero heatmaps rival learned ones—heatmap quality has limited leverage

- Uniform and near-zero priors can rival learned heatmaps under current MCTS designs.
- On small scales, giving more time mainly benefits LKH-3, shrinking relative gains from heatmaps.
- Overall leverage of heatmap quality is modest compared with search design and budget.

Findings may shift beyond Euclidean TSP and our implementation choices

- Results are on synthetic Euclidean instances; other metrics or data may differ.
- Hardware and engineering details influence wall-clock comparisons.
- Scope is a position with baselines and critiques, not a new end-to-end solver.

Theory-aligned objectives and end-to-end ML offer the path forward

- Design objectives that align heatmap learning with search-time performance.
- Close training–inference gaps using differentiable or faithful search proxies.
- Pursue autonomous, generalizable, end-to-end solvers beyond post-hoc search.

For practice today, choose LKH-3; if using MCTS, default to SoftDist

- For large-scale TSP under matched resources, prefer LKH-3 for reliability and speed.
- If using heatmap-guided MCTS, SoftDist is a strong default with negligible latency.
- Always weigh heatmap generation time against search time in end-to-end budgets.

Open questions: objectives, fairness, and generalization

- What objectives best align learning with the deployed search?
- How should we enforce resource fairness across diverse hardware stacks?
- Can end-to-end ML solvers generalize beyond Euclidean assumptions?

Acknowledgments

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